

Model Explanation using Influence Analysis

Project for Data Centric AI course

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Project repository: <https://github.com/aledigirm3/advanced-topics-computer-science/tree/main/DataCentric-AI>

Keywords: LLM, Computer Science, Model Explanation, Data Centric AI, First-Order, TracIn

1. Introduction

In the context of Artificial Intelligence and Machine Learning, model interpretability is becoming increasingly central. In particular, understanding why a model makes a given decision is crucial not only for ethical and regulatory reasons but also to enhance users' trust in automated predictions. One of the most promising directions for model interpretability is the investigation of how training data influence model behavior.

Influence analysis methods, such as First-Order Influence Functions and TracIn, aim to estimate the extent to which individual training samples contribute to a model's prediction on a specific test point.

However, a still underexplored yet fundamentally important aspect is the stability and consistency of such explanations: can we trust the outcomes of these methods? Do they yield similar results when applied to the same problem? Most importantly, are they capable of identifying truly influential samples whose removal significantly degrades model performance?

This study situates itself within this context and aims to empirically evaluate the internal consistency and external validity of explanations generated by First-Order Influence and TracIn, using a relatively simple yet well-known tabular classification task: predicting the survival of Titanic passengers. The use of a standardized dataset allows us to focus on the methodological aspects of influence analysis, minimizing the confounding factors related to domain or model complexity.

In the following sections, we describe the experimental design, report both quantitative and qualitative results, and discuss the implications of our findings in terms of the reliability and practical utility of influence-based explanation methods.

For a more detailed overview of the experiment, including all steps and visualizations, please refer to the Jupyter notebook *model_explanation.ipynb*.

2. Experimental Design

The objective of this study was to analyze the stability and consistency of influence-based explanation methods (First-Order Influence and TracIn) on a tabular classification task (Titanic dataset).

1. **Model and Data:** A sequential neural network implemented with TensorFlow/Keras was trained on the Titanic dataset, following careful preprocessing to handle missing values and categorical features.
2. **Distribution Analysis:** To examine the nature of the influence score rankings, we plotted the full distribution of influence scores for a selected test sample, verifying the presence of long-tail effects.
3. **Consistency:** To quantify the agreement between the two methods, we computed Kendall's Tau rank correlation between the influencer lists generated for the same five test samples.
4. **Validity/Stability:** To validate the effectiveness of the explanations, we conducted a removal experiment: the top 20 influencers identified by TracIn were removed from the training set, and a new model was trained from scratch to measure the impact on performance (Accuracy, Loss).
TracIN has been chosen to identify the top 20 influencers for our usual test sample (it is generally lighter since it doesn't directly compute the Hessian, and it is also less affected by outliers).

3. Results

1. **Score Distribution:** Both methods exhibited a long-tailed distribution (especially First-Order), where a small number of training samples had very high influence, while the majority had negligible impact.
See the graphs on jupyter file.
2. **Consistency (Kendall's Tau):** The consistency scores between First-Order and TracIn were generally low (typically ranging from 0.1 to 0.3), indicating a significant divergence in the rankings produced by the two methods.
3. **Qualitative Analysis:** Manual inspection of the top influencers confirmed this disagreement, as the two methods frequently identified different passengers as the

most “important”, which aligns with the previously calculated Kendall Tau correlation (which already gave us discriminative probability values supporting this observation).

4. **Impact of Removal (with TracIn):** Removing the top 20 influencers led to a measurable decline in the performance of the retrained model (decreased accuracy and increased loss), validating that TracIn was able to identify training samples that were genuinely impactful for the learning process.

	Accuracy	Loss
Original Training Set	0.8202	0.4986
New Training Set	0.7416	0.5265

Table 1: Performance metrics after removal of influential samples

4. Conclusion

This study highlights two critical aspects of influence-based explanation methods:

- **Fragility and Inconsistency:** Explanations are not absolute. Different methods yield different results, and therefore, one cannot blindly rely on a single “explanation.” The low consistency suggests that these methods capture different facets of how data influence the model.
- **Contextual Validity:** Despite their inconsistency, these methods appear capable of identifying relevant data samples, as demonstrated by the removal experiment. Their value does not lie in providing an “objective truth,” but rather in offering a powerful heuristic to uncover the most critical data points that shaped a specific model decision.

In conclusion, influence analysis techniques represent a valuable tool for interpretability, but they must be employed with an awareness of their sensitivity and limitations. Comparing multiple methods and validating their findings, as done in this assignment, is a crucial step toward robust analysis.